

TAPLite Kernel Computational-Efficiency Enhancements

Warm Starts, SP-Tree Sharing, and Column Stores

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July 2026

Headline: what this buys the looped 4-step process

Assignment runs **once per global feedback loop** (typically 4–8 loops $\times$ 5 periods = 20–40 calls); all entries from our measured numbers, derived rows labeled.

Lever	Factor	Basis
1. L1 within-loop warm start (shipped)	$\sim 45\%$ fewer iters (~ 11 vs 20); 1.9x tighter	Measured: SCAG AM L1 table
2. L2 loop-over-loop restart (shipped)	2.2x per call (35 s vs 78 s); starts at prior gap 0.072%	Measured: SCAG AM L2 table
3. Whole feedback run	$\sim 1.8x$ warm starts alone	Derived: rows 1–2
4. L3 column replay + GP sweeps (shipped)	139x to 0.10% gap; 8.4x to 0.05%; $\leq 0.02\%$ <i>only</i> via GP	Measured: SCAG same-gap table (compute-only)

Derived arithmetic: 5-loop run = $5T$ cold vs $T + 4 \times 0.45T \approx 2.8T \Rightarrow \sim 1.8x$ for the full feedback run. Multi-class runs save another 2–3x on SP via tree sharing (6 modes $\rightarrow$ 4 trees, byte-identical); all numbers sit on the 30/30 regression + byte-identity gate.

The kernel's shortest-path engine was modernized to industry-standard heap label-setting (21x on our prior internal baseline) — treated here as a prerequisite, not a claimed advance.

Motivation

- Regional assignment runs are rarely one-off: VDF recalibration, toll tweaks, demand scenarios, gap-tightening continuations.
- A cold Frank–Wolfe start rediscovers congestion every time: **SCAG RTP24 AM iteration-1 gap = 185%** from free-flow times.
- Goal: reuse what the previous run already computed — *without* ever compromising exactness or trust.
- Method: a structured **industry panel** of five expert personas, each reviewing the actual kernel source.

The industry-panel method

Five personas, five traditions, one kernel (TAPLite.cpp):

1. Caliper-class commercial vendor engineer
2. PTV/Visum-class commercial vendor engineer
3. Open-source lead (AequilibraE-class)
4. Bar-Gera-tradition academic (convergence theory)
5. HPC/systems engineer (memory layout, binary formats)

Each proposed accelerations with effort estimates and registered *dissents*.
The panel converged on one organizing artifact: the **warm-start ladder**.

The warm-start ladder L1–L4

- L1 – TIMES** Preload congested link times from a prior run. Feasibility untouched; equilibrium unchanged. All 5 endorse; ship first.
- L2 – FLOWS** Restore link volumes, skip iteration-0 AoN. Valid only for *identical demand* (fingerprint-enforced).
- L3 – COLUMNS** Per-OD path sets with θ shares: a *demand-invariant routing policy*, replayable against any new OD table. Unanimously “the strategic asset.”
- L4 – STATE** mmap-able full snapshot (DTST); kills the input-parse phase. Planned.

Universal rule: reset CFW/BFW direction history on any warm start.
Startup resolution: DTAC $\rightarrow$ L3, else DTLR $\rightarrow$ L2, else DTLP $\rightarrow$ L1, else free-flow — every decision logged.

P0 (shipped): the exact trio

All three verified **bit-identical** via byte-diffs of `link_performance.csv` + gap trajectories vs the pre-change exe.

- **L1 warm_start_times**: preload congested times from a prior `link_performance` CSV or DTLP binary, matched by external link id. Bad input $\Rightarrow$ loud warning, cold start; never fails.
- **SP-equivalence auto-detect**: modes share one tree iff `allowed_use` and per-link additional cost are identical everywhere. PCE/occupancy never split a group. 6 modes $\rightarrow$ 4 trees on Chicago Sketch multimodal, byte-identical; grouping always logged.
- **Group-aware zero-demand-origin skip**: per-(mode, origin) SP skip with direct BIGM fill — identical results by construction.

Regression: **30/30 checks pass, no re-baselining.**

P0 evidence: SCAG RTP24 AM, 224,288 links

	Cold	L1 warm start
Iteration-1 gap	185.2%	44.9%
lters to cold's final gap	19	~11
Final gap (iter 19)	0.0755%	0.0407%

- ~**1.9x tighter** final gap, ~2x fewer iterations to equal gap.
- 224,288 / 246,806 links matched (connectors keep free-flow time by design).
- Only the starting point changes — same equilibrium.

P1 (shipped): DTLR snapshot + warm_start_flows

- `flow_snapshot=1` writes `link_flows.bin` (magic DTLR): main + per-mode volumes by external link id.
- Header carries a **16-byte demand fingerprint**: OD total, positive-cell count, min/max zone id — deterministic accumulation.
- On match: restore volumes, recompute times, *skip iteration-0 AoN*, continue FW from iteration 1. The restored point is a convex combination of AoN assignments of the same demand $\Rightarrow$ a valid FW iterate.
- On mismatch (or wrong magic / mode count): **loud warning, both fingerprints printed, demote to L1** (times-only seed) — never a silent restart from incompatible flows.
- Guard verified: one OD cell perturbed 10% on Chicago Sketch flips the fingerprint and triggers the demotion.

P1 evidence: SCAG restart continuation

	Run A: cold, 20 it	Run B: warm, 10 it
Links restored	—	246,806 / 246,806
Iteration-1 gap	185.23%	0.0724%
Final gap	0.0755%	0.0453%
FW wall time	78 s	35 s

- Run B starts *at* run A's final gap — a seamless continuation.
- **1.7x tighter** gap in **0.45x** the wall time.
- Identical-demand reruns (recalibration, gap tightening) lose no work.

P2 increment 1 (shipped): DTAC column store

column_output=1 writes route_columns.bin (magic DTAC), sparse CSR over positive-demand OD pairs: the *last-iteration shortest path* per (mode, origin, destination) as external-link-id sequences.

Chicago Sketch (20 it, 3 modes, 387 zones)

Stored paths / link entries	93,135 / 1,211
DTAC vs route_assignment.csv	5.6 MB vs 55.1 MB (9.8x smaller)
Push-down R^2 vs link volumes	0.966 vs 0.966
Turn-restriction single-path net	$R^2 = 1.0$, banned movement avoided

Honest caveat: R^2 0.966 is expected — link volumes are the FW blend of 20 iterations; increment 1 stores only the final AoN direction. Exactness arrives with increment 2's θ shares.

P2 increment 2 / full L3 (shipped): θ shares, replay, GP

- **DTAC v2 θ -share columns:** the path SET across all FW iterations, exact FW cascade $\Rightarrow$ push-down $R^2 = 1.000000$, max diff 0.00 veh (increment 1: 0.9657).
- **warm_start_columns:** scatter $\theta \times$ the *current* OD table — demand *may* differ (that is the point); paths broken by network edits dropped + θ renormalized; skip iteration-0 AoN, resume FW.
- **column_adjust_sweeps:** fixed-policy GP sweeps (Gauss–Seidel, exact VDF derivatives, no SP calls).
- **Mandatory two-gap report:** *restricted* gap (stored paths) and *TRUE* gap (fresh SP) always printed — restricted never masquerades as UE. Freeze/replay = 0 FW iterations + same report.

Regression 30/30; defaults-off `link_performance.csv` AND `route_assignment.csv` byte-identical to the pre-change exe.

L3 evidence: same-gap trade-off, SCAG RTP24 AM

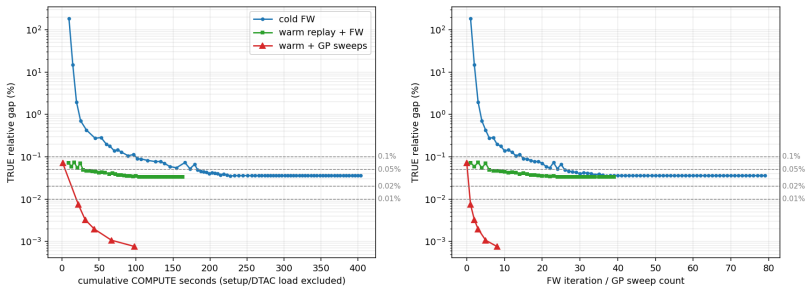
246,806 links, 342,712 OD cells; demand perturbed +5% on 1,000 cells. **Compute seconds** to a TRUE-gap target (setup + DTAC load excluded — the accounting rule; kernel prints phase timings).

Target	Cold FW	Warm replay+FW	Warm + GP	Speedup
0.10%	101.7 s @ 15 it	8.6 s @ 1 it	0.7 s @ scatter	139x
0.05%	183.4 s @ 26 it	28.5 s @ 6 it	21.9 s @ 1 sweep	8.4x
0.02%	not reached (0.036%)	not reached (0.033%)	21.9 s @ 1 sweep	only GP
0.01%	not reached	not reached	21.9 s @ 1 sweep	only GP

The regime split: plain FW — cold OR warm-started — plateaus above $\sim 0.03\%$; GP sweeps over the stored columns pass 0.01% in 1–3 sweeps and keep going (8 sweeps $\rightarrow 7.6 \times 10^{-4} \%$). The sweeps reach targets FW cannot. Chicago Sketch: 4.8x @ 0.10%, 2.4x @ 0.05%, GP-only below.

L3 evidence: the trade-off figure

SCAG RTP24 AM (246,806 links, 342,712 OD cells): perturbed demand (+5% x 1,000 OD cells) -- same-gap trade-off
cold FW & warm+FW traces = per-iteration TRUE gap; GP-sweep points = final TRUE gap per sweeps= k replay run (sweep-time axis excludes the fresh-SP gap audit)



FW curves = per-iteration TRUE gap; GP points = final TRUE gap of each sweeps= k replay run.
Companion: 13_tradeoff_sketch.png (Chicago Sketch, same regimes).

The standard rerun recipe (team decision)

Rerunning ANY model = L3 warm start to a 0.1% gap

`warm_start_columns = route_columns.bin`

`convergence_gap_pct = 0.1`

`column_adjust_sweeps = 0`

- Baseline run sets `column_output=1` once.
- Replay+FW reaches the 0.1% TRUE gap in ~ 1 iteration — **139x** less compute than cold on SCAG.
- Add sweeps only for tighter targets (1 sweep $\rightarrow$ 0.01% on SCAG).
- Kernel cold-start defaults unchanged; settings-only, opt-in.
- Full recipe + guard behavior:
`github_taplite/docs/RERUN_RECIPE.md`.

The binary format family

Magic	Role	Written by	Consumed by
DTAB	OD demand	demand-bin tool	demand_format=1 (2.5–3x load)
DTLP	Link performance	link_output=3	L1 warm start; reporting
DTLR	Flow restart	flow_snapshot=1	L2 warm_start_flows
DTAC	Path columns (v2: θ)	column_output=1	verifier; L3 warm_start_columns
DTST	Full state	planned (L4)	planned: mmap, skips input path

Common contract: magic + version headers, external-link-id matching, loud fallback on invalid input — a run is never aborted by an optimization hint.

Engineering principles

- **Exact or clearly reported** — bit-exact features, or a printed, quantified deviation (DTAC R^2 ; restricted-vs-full gap).
- **Default-off** — every behavior-changing key ships off and is registered in the dtalite_qa schema.
- **Loud-warn, never fail** — bad warm-start input degrades gracefully (cold start or L1 demotion), with match rates logged.
- **Fingerprint guards** — one perturbed OD cell flips the demand fingerprint and demotes the restart.
- **Regression + byte-identity gates** — 13 networks / 30 checks, no re-baselining, byte-diffs vs the pre-change exe.
- **No bush-based rewrite** — unanimously rejected; GP polish on stored columns gives “bush-method-class precision without becoming Algorithm B.”

Summary

- **P0 shipped:** L1 times warm start + SP-tree sharing + origin skip — all exact, 30/30 regression.
- **P1 shipped:** DTLR flow restart with demand-fingerprint guard — SCAG restarts *at* the prior final gap; 1.7x tighter in 0.45x time.
- **P2/L3 shipped:** DTAC v2 θ -share columns (push-down $R^2 = 1.000000$), rescale-to-new-OD replay, GP sweeps, freeze/replay with the mandatory two-gap report.
- **Same-gap headline (SCAG):** 139x @ 0.10%, 8.4x @ 0.05%; $\leq 0.02\%$ only via GP (FW plateaus $\sim 0.03\%$; GP $\rightarrow 8 \times 10^{-4} \%$).
- **Standard rerun recipe:** warm_start_columns + 0.1% gap target + 0 sweeps — every rerun, any demand.
- **Remaining:** per-origin parallel GP sweeps; L4 DTST state snapshot (kill the setup phase).
- The ladder compounds: every rung reuses the formats and guards of the one below it.